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Comparative Analysis of Image Classification Model on MNIST and CIFAR-10

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Introduction

- **Motivation for Image Classification**
  - Image classification serves as a fundamental task in computer vision
  - The ability to automatically categorize images into predefined classes is a precursor to more complex tasks such as object detection, semantic segmentation, and autonomous navigation
  - As neural network architectures evolve, it becomes critical to rigorously evaluate their performance characteristics across datasets of varying complexity.
- **Objective**
  - To systematically compare Classical Machine Learning (Logistic Regression, SVM, kNN) against Deep Learning models.
  - To quantify the impact of inductive biases (such as translation invariance in CNNs) on performance.
  - To analyze the relationship between model depth and dataset complexity by benchmarking on MNIST and CIFAR-10.

Datasets

MNIST (Modified National Institute of Standards and Technology)

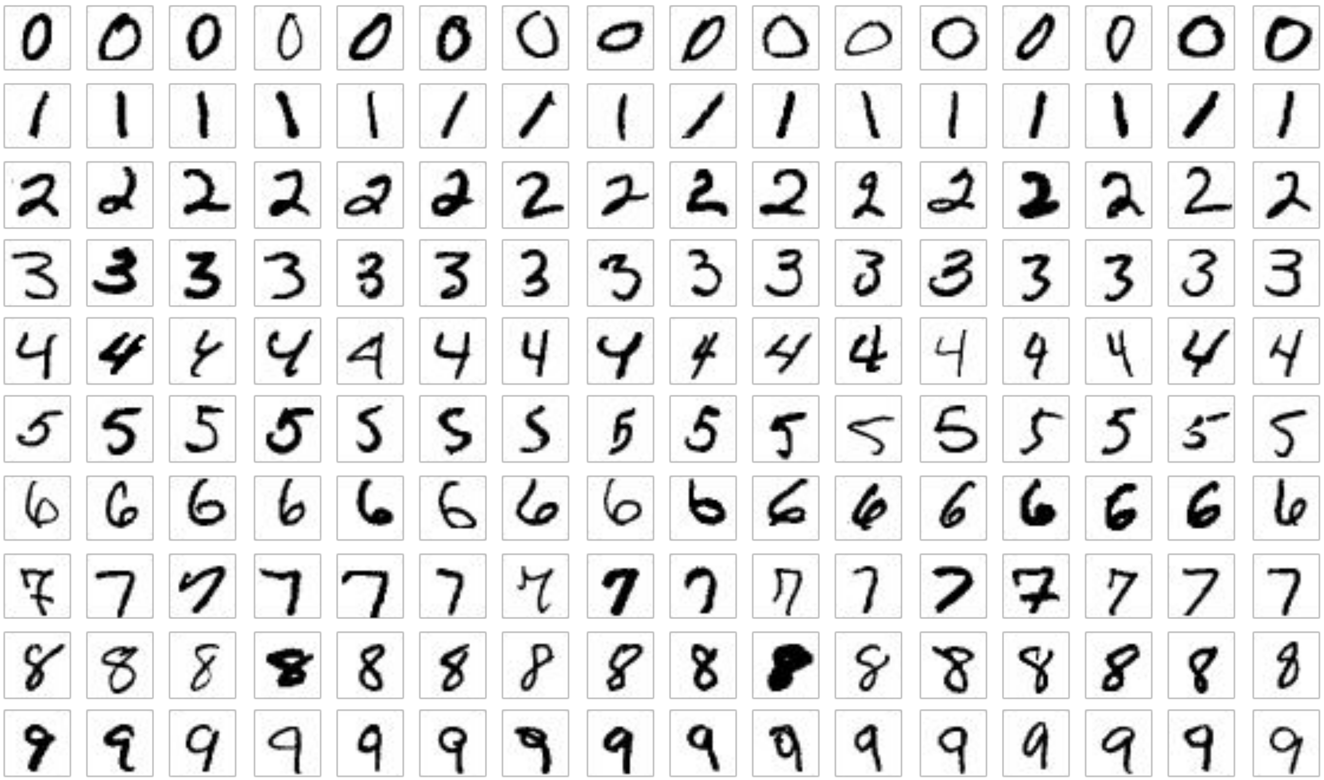
- **Characteristics:** The dataset consists of 60,000 training and 10,000 test images of handwritten digits (0–9). Each image is a 28 x 28 grayscale grid.
- **Suitability:** MNIST represents a "low-complexity" structural task. The digits are centered and lack background noise. It serves as an initial sanity check; models that fail to converge on MNIST are fundamentally flawed. However, its simplicity often masks the limitations of shallow architectures.

CIFAR-10 (Canadian Institute for Advanced Research)

- **Characteristics:** This dataset contains 60,000 training and 10,000 test images across 10 classes (e.g., Airplane, Automobile, Bird, Cat). Images are 32 x 32 pixels with 3 color channels (RGB).
- **Suitability:** CIFAR-10 introduces significant "high-complexity" challenges, including background clutter, varying lighting conditions, object occlusions, and intra-class variability (e.g., a bird can appear in many poses and colors). It serves as a stress test for model generalization.

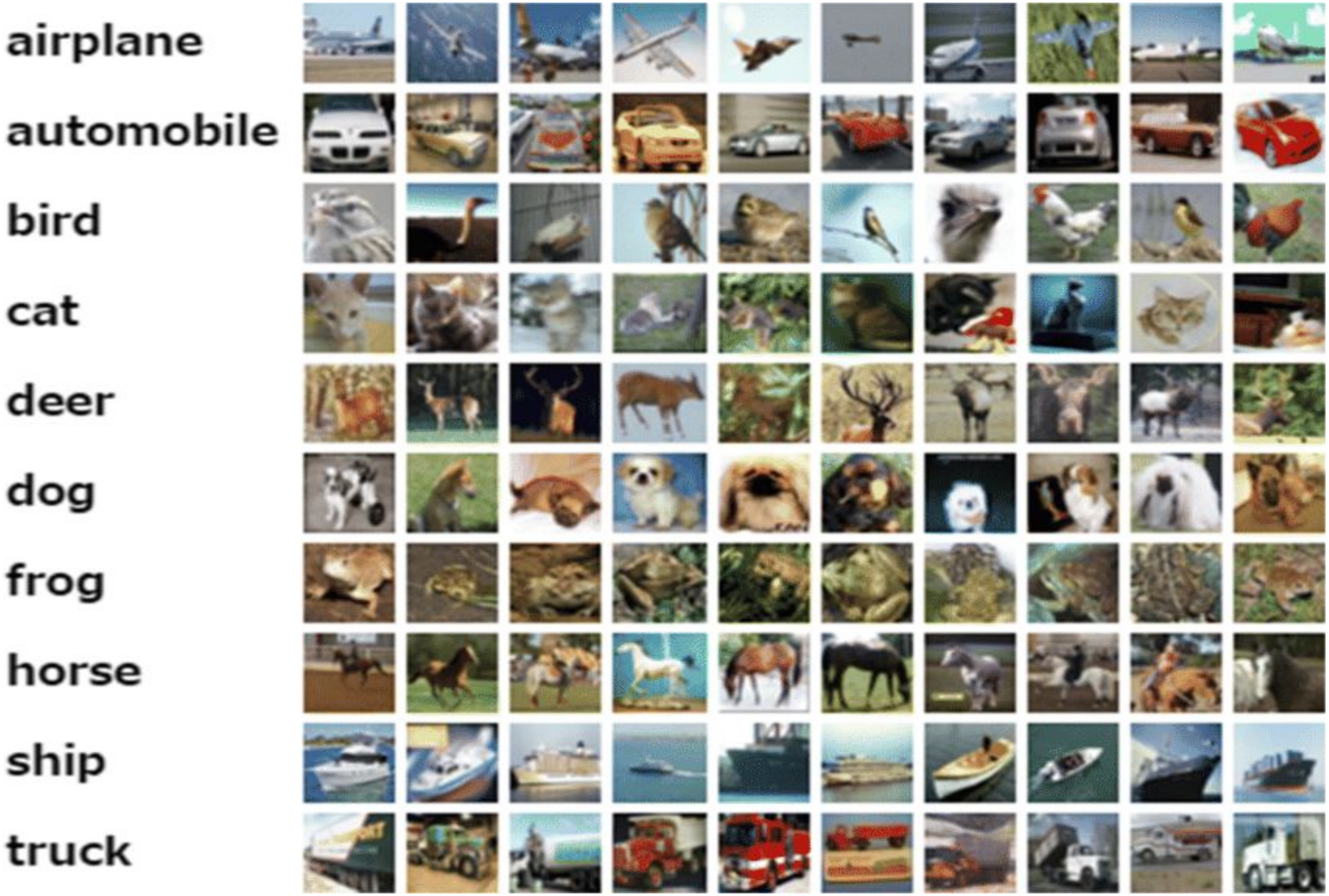
Datasets

MNIST



Datasets

CIFAR-10



Experimental Pipeline

- **Data Loading and Preprocessing**
  - **Normalization:** All images were transformed to tensors and normalized. For MNIST, we standardized inputs using mean=0.1307 and std=0.3081. For CIFAR-10, we used channel-wise means and standard deviations to accelerate gradient convergence.
  - **Flattening:** For classical models (SVM, MLP) that do not support 2D inputs, images were flattened into 1D vectors (784 features for MNIST, 3072 for CIFAR-10).
- **Data Splitting**
  - **Training Set:** Used for parameter optimization.
  - **Validation Set (10%)**: A held-out subset of the training data used strictly for hyperparameter tuning and early stopping.
  - **Test Set**: Used only for the final performance evaluation.
- **Augmentation Strategy**
  - **CIFAR-10**: To combat overfitting in deep networks, we applied RandomCrop and RandomHorizontalFlip during training. This forces the model to learn invariant features rather than memorizing static pixel arrangements.
  - **MNIST**: No augmentation was applied. The digits are spatially aligned, and augmentation (like rotation) could inadvertently change the class semantics (e.g., rotating a '6' to a '9').

Model Categories and Scenarios

Scenario A: Classical Machine Learning vs. Deep Learning (MNIST Only)

We evaluated **Logistic Regression**, **Support Vector Machines (SVM)**, and **k-Nearest Neighbors (kNN)** on MNIST.

- **Justification:** These models provide a baseline for computational cost and accuracy. They treat images as unstructured vectors, ignoring spatial relationships.
- **Exclusion from CIFAR-10:** These models were excluded from the CIFAR-10 benchmark due to the "Curse of Dimensionality." The pixel-wise Euclidean distance used by kNN and SVM becomes semantically meaningless in high-dimensional RGB space.

Model Categories and Scenarios

Scenario B: Model Capacity Comparison

We compared three neural architectures:

1. **MLP (Multilayer Perceptron):** A dense network with no spatial inductive bias.
2. **Simple CNN:** A shallow network (2 convolutional layers) designed to test the benefit of local feature extraction.
3. **ResNet-18 (Deep CNN):** A deep residual network designed to test the benefit of hierarchical feature learning.

Scenario C: Dataset Difficulty Comparison

By applying the **Simple CNN** to both datasets, we isolated the impact of data complexity. The performance drop from MNIST to CIFAR-10 using the exact same architecture quantifies the increased difficulty of the RGB object recognition task.

Training and Evaluation Strategy

Training Design

- **Optimization:** The **Adam** optimizer was selected for its adaptive learning rate capabilities, providing faster convergence than standard SGD.
- **Early Stopping:** Training was halted if validation loss did not improve for 5 consecutive epochs, preventing overfitting and saving computational resources.

Metrics

- **Accuracy:** The primary metric for overall performance.
- **F1-Score (Macro):** Calculated to ensure that performance is balanced across all classes, preventing bias towards easier classes.
- **Confusion Matrix:** Used for qualitative error analysis to identify specific class confusion pairs.

Configuration

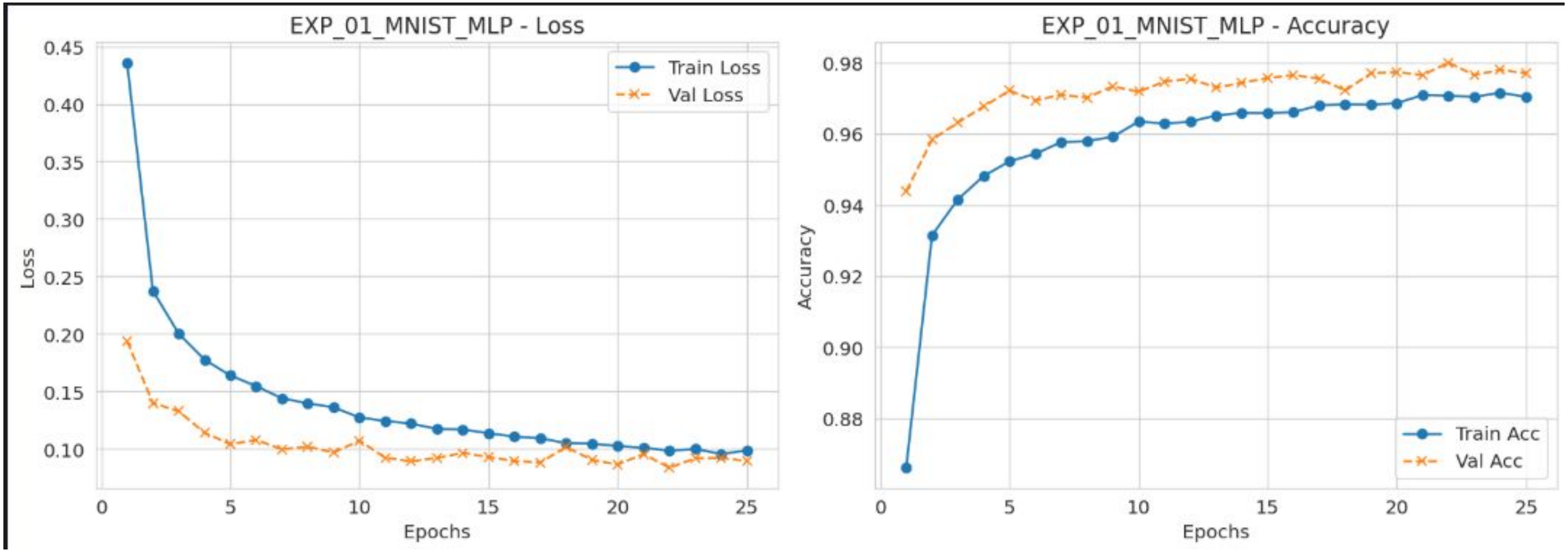
Model	Key Hyperparameters
Logistic Regression	solver=lbfgs, max_iter=1000
k-Nearest Neighbors	n_neighbors=5, metric=minkowski
Support Vector Machine (SVM)	kernel=rbf, $C = 1.0$, gamma=scale

Parameter	EXP_01 (MLP)	EXP_02 (CNN)	EXP_03 (CNN)	EXP_04 (ResNet-18
Dataset	MNIST	MNIST	CIFAR-10	CIFAR-10
Input Shape	$1 \times 28 \times 28$	$1 \times 28 \times 28$	$3 \times 32 \times 32$	$3 \times 32 \times 32$
Batch Size	64	64	64	128
Learning Rate	0.001 (10^{-3})	0.001	0.001	0.001
Max Epochs	15	10	20	25
Dropout Rate	$p = 0.5$	$p = 0.25$	$p = 0.25$	None (Batch Normalizati

Evaluation

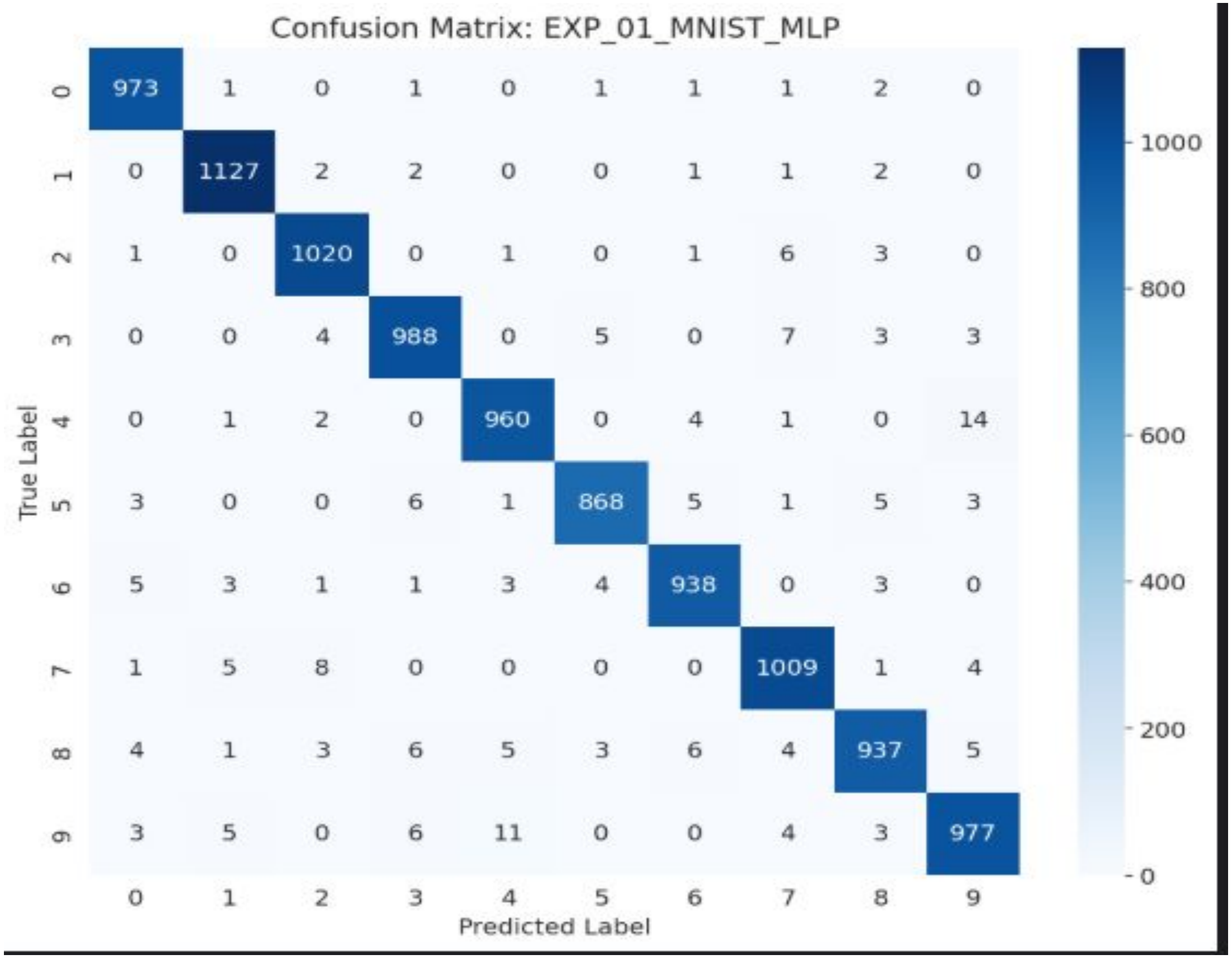
Category	Model	Dataset	Accuracy	F1-score
Classical ML	Logistic Regression	MNIST	0.9228	0.9217
Classical ML	k-NN	MNIST	0.9680	0.9679
Classical ML	SVM (RBF)	MNIST	0.9783	0.9782
Deep Learning	MLP	MNIST	0.9797	0.9795
Deep Learning	Simple CNN	MNIST	0.9890	0.9889
Deep Learning	Simple CNN	CIFAR-10	0.7686	0.7680
Deep Learning	ResNet-18	CIFAR-10	0.8757	0.8762

Visualize

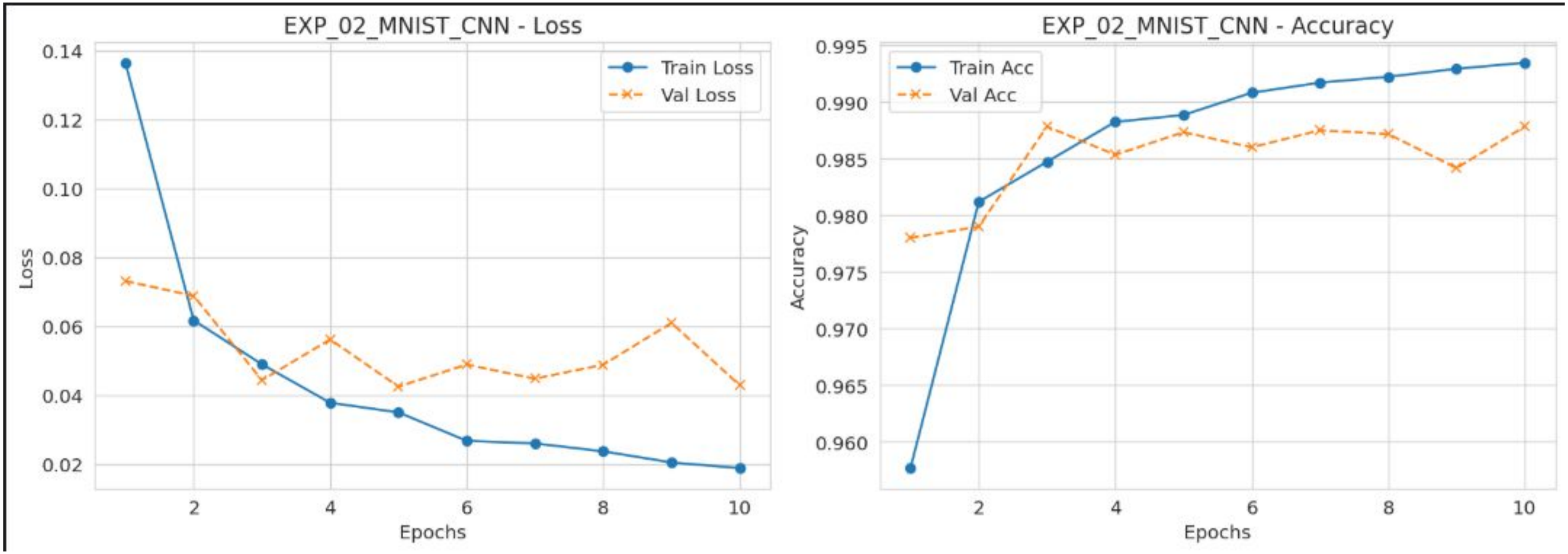


EXP_01_MNIST_MLP - Loss
Train Loss
Val Loss
Loss
Epochs
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0
5
10
15
20
25
EXP_01_MNIST_MLP - Accuracy
Train Acc
Val Acc
Accuracy
Epochs
0.98
0.96
0.94
0.92
0.90
0.88
0
5
10
15
20
25

Heatmap

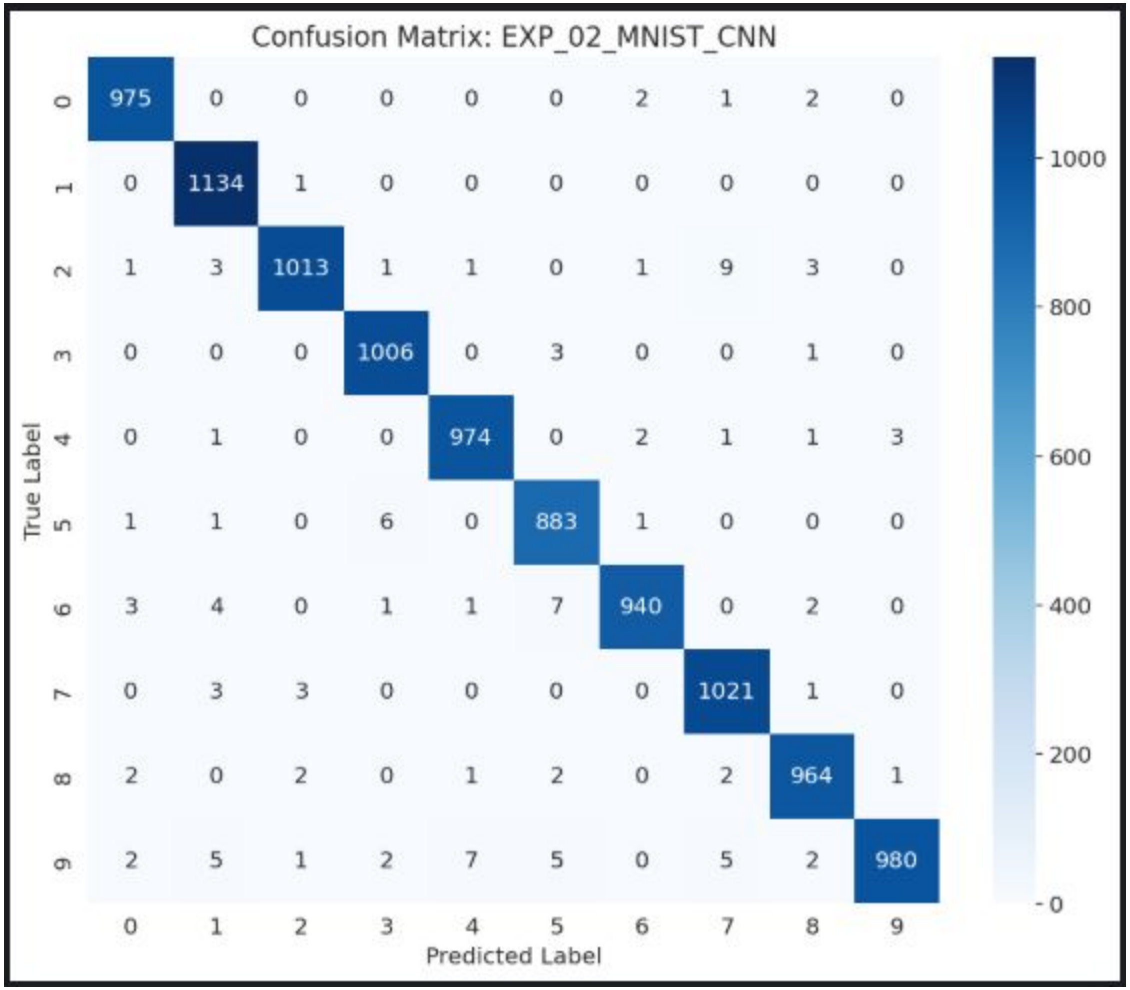

Confusion Matrix: EXP_01_MNIST_MLP
True Label
Predicted Label
0 1 2 3 4 5 6 7 8 9
973 1 0 1 0 1 1 1 2 0
0 1127 2 2 0 0 1 1 2 0
1 0 1020 0 1 0 1 6 3 0
0 0 4 988 0 5 0 7 3 3
0 1 2 0 960 0 4 1 0 14
3 0 0 6 1 868 5 1 5 3
5 3 1 1 3 4 938 0 3 0
1 5 8 0 0 0 0 1009 1 4
4 1 3 6 5 3 6 4 937 5
3 5 0 6 11 0 0 4 3 977
1000
800
600
400
200
0

Visualize

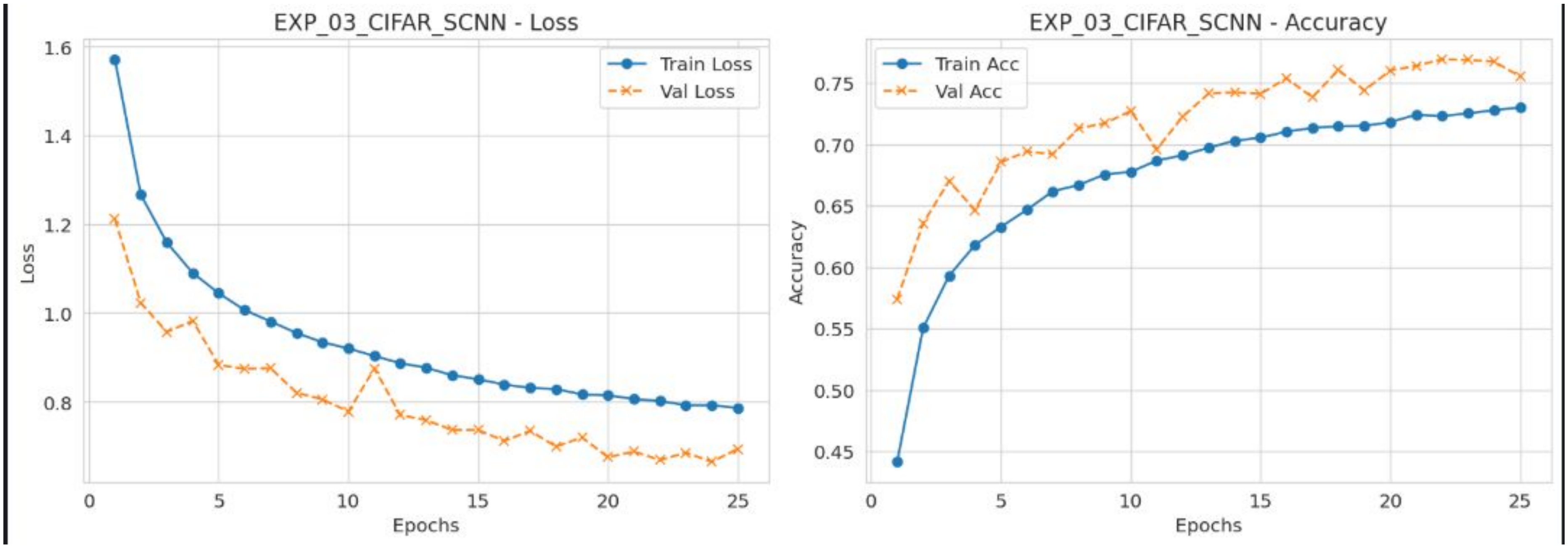


EXP_02_MNIST_CNN - Loss
Train Loss
Val Loss
Loss
Epochs
0.14
0.12
0.10
0.08
0.06
0.04
0.02
2
4
6
8
10
EXP_02_MNIST_CNN - Accuracy
Train Acc
Val Acc
Accuracy
Epochs
0.995
0.990
0.985
0.980
0.975
0.970
0.965
0.960
2
4
6
8
10

Heatmap

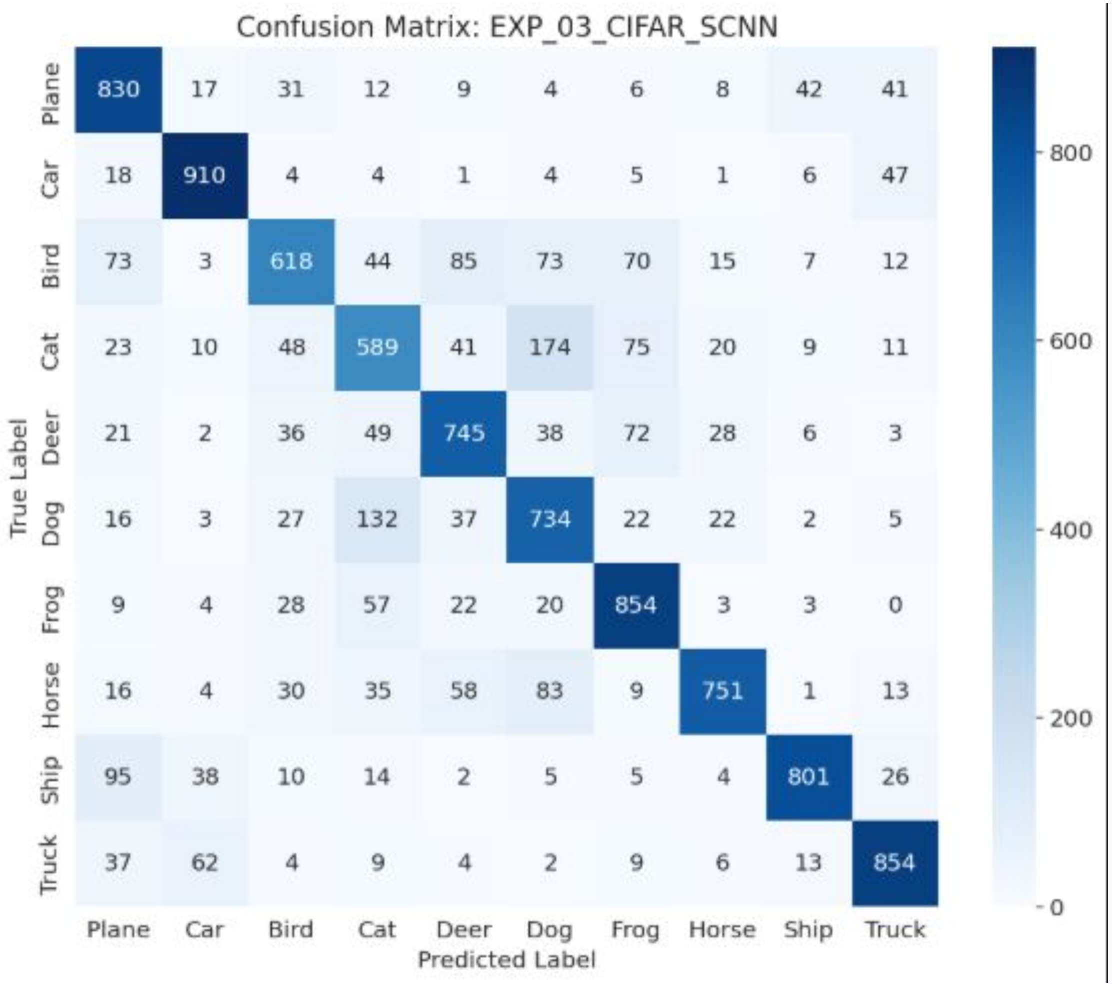

Confusion Matrix: EXP_02_MNIST_CNN
True Label
Predicted Label
0 975 0 0 0 0 0 2 1 2 0
1 0 1134 1 0 0 0 0 0 0 0
2 1 3 1013 1 1 0 1 9 3 0
3 0 0 0 1006 0 3 0 0 1 0
4 0 1 0 0 974 0 2 1 1 3
5 1 1 0 6 0 883 1 0 0 0
6 3 4 0 1 1 7 940 0 2 0
7 0 3 3 0 0 0 0 1021 1 0
8 2 0 2 0 1 2 0 2 964 1
9 2 5 1 2 7 5 0 5 2 980
0 1 2 3 4 5 6 7 8 9
1000
800
600
400
200
0

Visualize

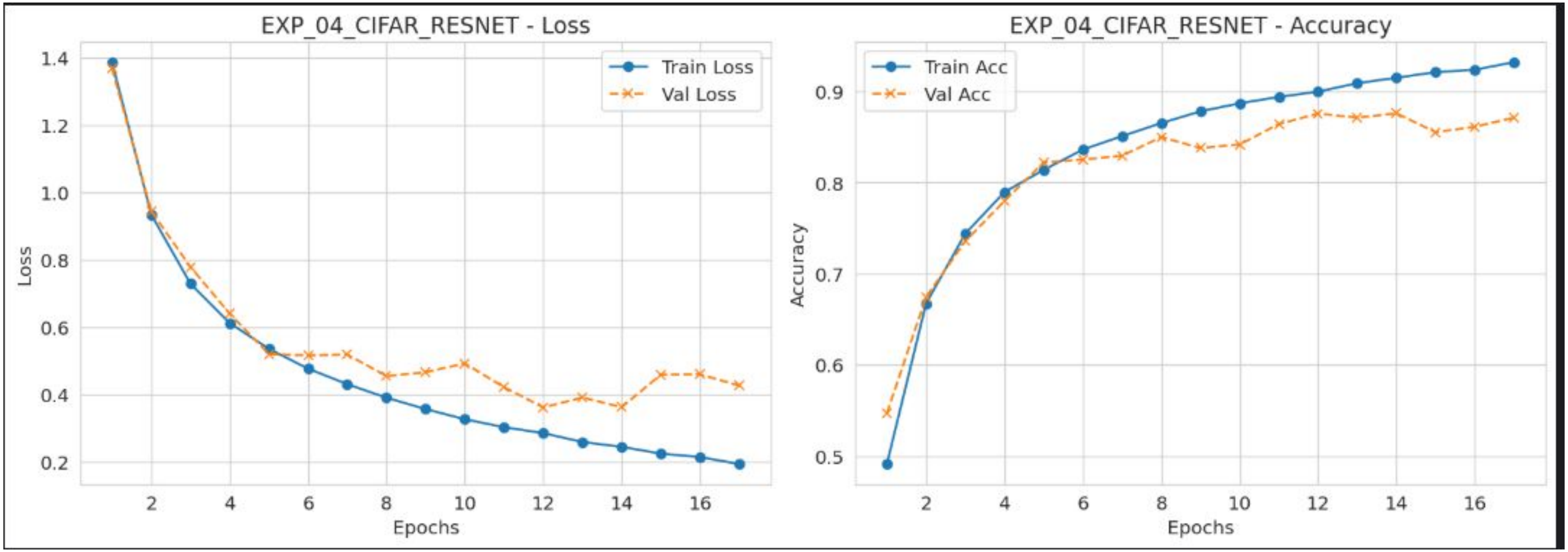


EXP_03_CIFAR_SCNN - Loss
Train Loss
Val Loss
Loss
Epochs
1.6
1.4
1.2
1.0
0.8
0
5
10
15
20
25
EXP_03_CIFAR_SCNN - Accuracy
Train Acc
Val Acc
Accuracy
Epochs
0.75
0.70
0.65
0.60
0.55
0.50
0.45

Heatmap

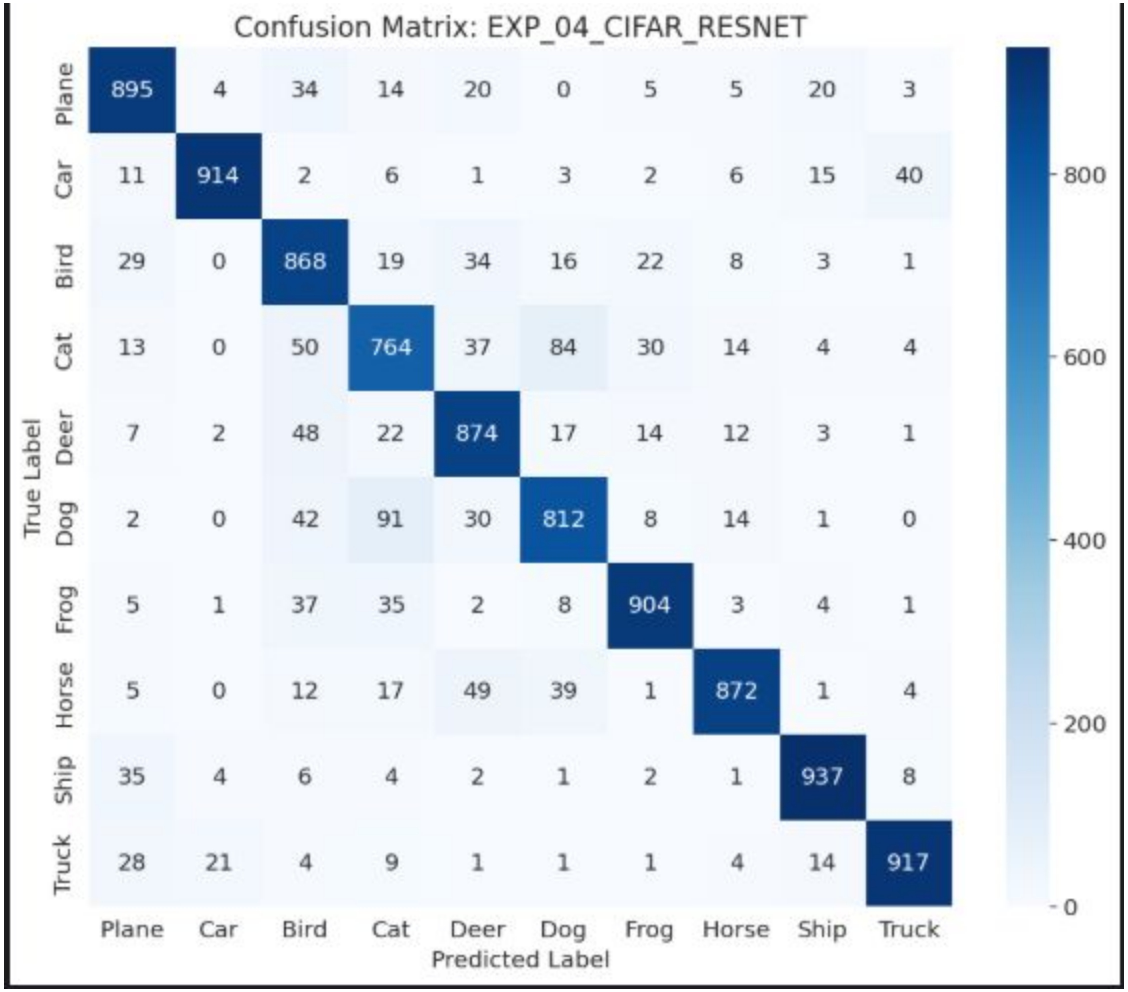

Confusion Matrix: EXP_03_CIFAR_SCNN
True Label
Predicted Label
Plane
Car
Bird
Cat
Deer
Dog
Frog
Horse
Ship
Truck
830 17 31 12 9 4 6 8 42 41
18 910 4 4 1 4 5 1 6 47
73 3 618 44 85 73 70 15 7 12
23 10 48 589 41 174 75 20 9 11
21 2 36 49 745 38 72 28 6 3
16 3 27 132 37 734 22 22 2 5
9 4 28 57 22 20 854 3 3 0
16 4 30 35 58 83 9 751 1 13
95 38 10 14 2 5 5 4 801 26
37 62 4 9 4 2 9 6 13 854
800
600
400
200
0

Visualize



EXP_04_CIFAR_RESNET - Loss
Train Loss
Val Loss
Loss
Epochs
1.4
1.2
1.0
0.8
0.6
0.4
0.2
2
4
6
8
10
12
14
16
EXP_04_CIFAR_RESNET - Accuracy
Train Acc
Val Acc
Accuracy
Epochs
0.9
0.8
0.7
0.6
0.5
2
4
6
8
10
12
14
16

Heatmap


Confusion Matrix: EXP_04_CIFAR_RESNET
True Label
Predicted Label
Plane Car Bird Cat Deer Dog Frog Horse Ship Truck
Plane 895 4 34 14 20 0 5 5 20 3
Car 11 914 2 6 1 3 2 6 15 40
Bird 29 0 868 19 34 16 22 8 3 1
Cat 13 0 50 764 37 84 30 14 4 4
Deer 7 2 48 22 874 17 14 12 3 1
Dog 2 0 42 91 30 812 8 14 1 0
Frog 5 1 37 35 2 8 904 3 4 1
Horse 5 0 12 17 49 39 1 872 1 4
Ship 35 4 6 4 2 1 2 1 937 8
Truck 28 21 4 9 1 1 1 4 14 917
800
600
400
200
0

THANK YOU !

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